FACULTY OF ENGINEERING, BUILT ENVIRONMENT, AND INFORMATION TECHNOLOGY (FOEBEIT)

BACHELOR OF COMPUTER SCIENCE

BCC1313 - ALGORITHMS & COMPLEXITY

FINAL ASSESSMENT

Customer Service Queue Simulation

INSTRUCTIONS TO CANDIDATE:

1. This is an **INDIVIDUAL** assignment.
2. This assignment contributes **40%** to your final course mark.
3. Compile all required files (report, source code, and screenshots) into **a single ZIP file** before submission.
4. Submit the **ZIP file link** via **SPARK by 11:59 PM on 6 December 2026.**

WARNING

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations. Plagiarism, paraphrasing and downloading large amounts of information from external sources will not be tolerated and will be dealt with severely.

SEGi University, Kota Damansara takes allegations of plagiarism very seriously. Submissions involving plagiarism will be marked but given zero mark. Plagiarism is the attempt to pass off the work of another as your own. Information taken from the work of others should be acknowledged by reference to obviate the charge of copying.

The work must be entirely your own. You will receive ZERO (0) for work that is not your own. The safety of your assessments is your responsibility. You must not permit another student access to your work.

Objective

This assignment assesses your ability to design and implement a Queue-based application in a real-world context. You will simulate a customer service system using C++/Java/Python, focusing on **Queue Abstract Data Type (ADT)**. The program should demonstrate basic queue operations and may optionally include priority handling for special cases.

Part A: Program Implementation

You are required to develop a **menu-driven console application that simulates a customer queue system**. The system should allow customers to join the queue, be served in order, and display the current queue status.

Choose **ONE (1)** of the following domains for the simulation:

1. Clinic or hospital waiting list
2. Fast food ordering system
3. Ticket counter or reservation system
4. Customer service centre

Functional Requirements

1. **Queue Management**

   The system should support the following basic queue operations:

   - Add a customer to the queue (collect details - name, age, gender, service type)
   - Serve the next customer in queue
   - Display the current queue
   - Exit the system

2. **Priority Handling *(optional)***

   To enhance the simulation, the system may include priority handling:

   - Customers who are senior citizens (age ≥ 60) or pregnant women should be given higher priority in the service order
   - Priority handling may be implemented using one of the following approaches:
     - Two separate queues (priority queue and normal queue)
     - A single queue with modified insertion logic

3. **Menu-Driven Interface**

   Provide a clear and user-friendly menu interface. An example menu is shown below:

```
Customer Service Queue System
1. Add New Customer
2. Serve Next Customer
3. Display Current Queue
4. Exit
```

Part B: Report

After completing your program, submit a **report** that includes both technical documentation and a self-reflective component.

I. Technical Documentation

Your report should include the following sections:

i) Introduction – Briefly describe your selected real-world scenario and the objective of the system.

ii) System Design – Explain the program structure, logic flow, and data handling.

iii) Queue Implementation – Describe how queue operations are handled.

iv) Optional Priority Logic – Explain how priority is implemented (*if included*).

v) Screenshots – Provide sample screeshots of your program's output.

II. Self-Reflective Component

Students are required to write a reflection addressing the following aspects:

A. Understanding and Approach

i) Explanation of the reasons for selecting the chosen scenario.

ii) Description of the approach used in system design and planning.

iii) Discussion of the challenges encountered and the strategies used to overcome them

B. Execution and Debugging

i) Explanation of how the correctness of the program was tested

ii) Description of the bugs or issues faced during development and how they were resolved.

C. Learning and Insights

i) Reflection on the understanding gained about the Queue ADT and programming concepts through this assignment.

ii) Explanation of how this assignment improved coding and problem-solving skills.

D. Application and Real-World Relevance

i) Explanation of real-world applications of queue systems.

ii) Discussion on possible extensions or improvements to the developed system.

Evaluation Criteria

Your work will be graded based on the following:

Criteria	Description	Marks (%)
i) Queue Functionality & Logic	Correct and complete implementation of queue operations	20%
ii) Menu Interface & Usability	Clear, functional, and user-friendly interaction interface	20%
iii) Optional Priority Handling (if included)	Logical and effective implementation of priority queue mechanism	10%
iv) Code Readability & Structure	Well-organized, readable code with proper naming and comments	20%
v) Report Quality & Reflection	Detailed and thoughtful answers in the report	20%
vi) Program Output Evidence	Adequate screenshots demonstrating full functionality	10%

Submission Requirements

1. **Source Code:** Submit your .cpp or .java file.
2. **Screenshots:** Provide screenshots showing program execution for all required functionalities.
3. **Report:** (in PDF format)
    - Technical documentation
    - Self-reflective analysis
    - Embedded screenshots
4. **Submission Deadline:** Submit all required files via **SPARK** on or before the due date.